In this lesson we explore how to obtain the graph of f' from the graph of f.

1. Use the provided Desmos file to answer the following questions:
   `https://www.desmos.com/calculator/7yc4glot2t`

   (a) What is true about the function f when the slope of the tangent line is positive? Negative?

   (b) What is the slope of the tangent line at a local maximum or minimum of f?

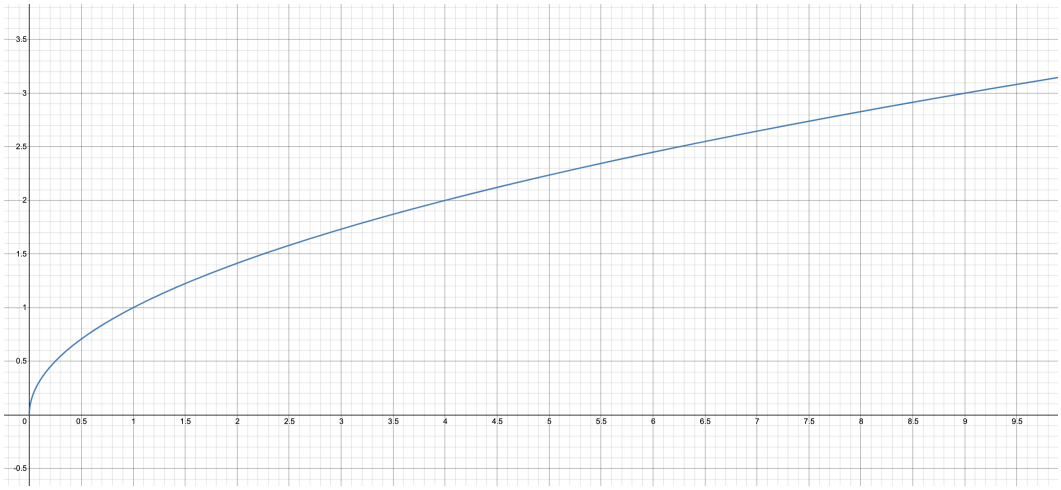
   (c) For $x < 0$, where is the tangent line relative to the graph of f? What about for $x > 0$?

   (d) What may be said about the shape of the graph of f for $x < 0$? What about for $x > 0$?

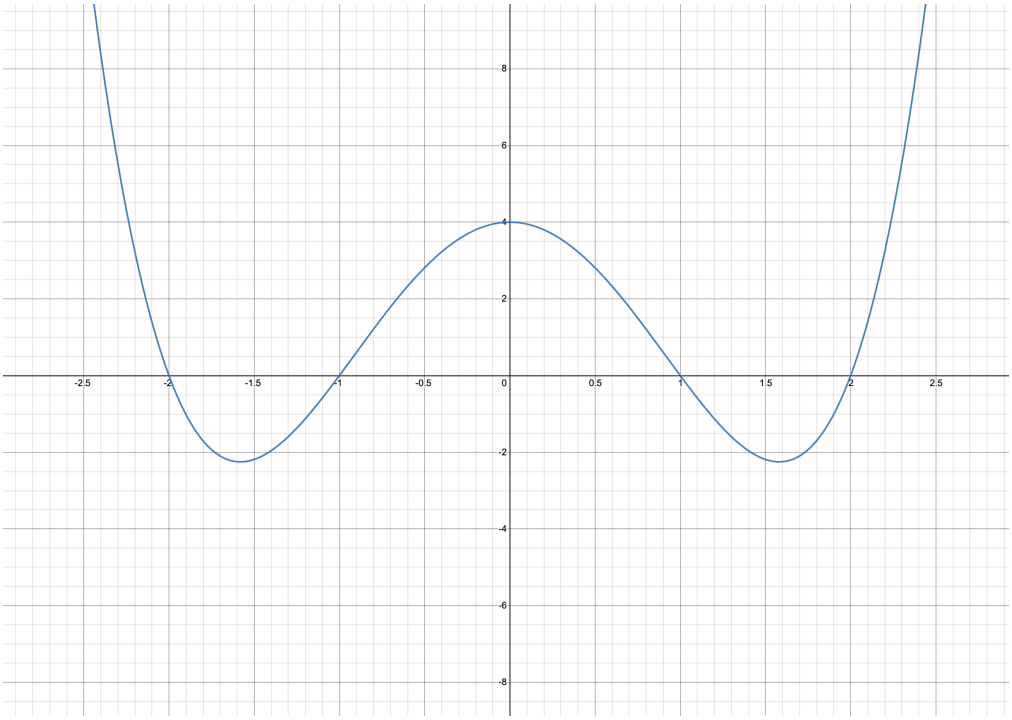
   (e) What happens to the tangent line at $x = 0$?

2. For each function graphed below, sketch the graph of its derivative by hand on the same axes.

(a)



(b)



Check your graphs with your instructor before moving on.

We now explore the relation between differentiability and continuity.

Theorem. If f is differentiable at a, then f is continuous at a.

3. Prove this theorem by completing the steps below.

   Proof. Suppose f is differentiable at a.

   (i) To show that f is continuous at a, we need to show that $\lim_{x\to a} f(x) =$ _____.

   (ii) Verify that, for $x \neq a$, $f(x)$ can be written as $f(x) = \frac{f(x)-f(a)}{x-a}(x-a)+f(a)$.

   (iii) Take the limit as $x \to a$ of both sides of the equation above and complete the proof.

4. Is the converse of the theorem above true? That is, if f is continuous at a, must f be differentiable at a? To investigate, consider the function $f(x) = |x|$. We've seen that $|x|$ is continuous everywhere. Does $f'(x)$ exist everywhere?

5. Suppose a function f is not continuous at $x = a$. *Could* it be differentiable at a?

Higher-Order Derivatives

The derivative of a function is itself a function, so we can find the derivative of the derivative.

Example. If $x = x(t)$ is the position of an object along a line at time t, then $\frac{dx}{dt}(t)$ is the *velocity* of the object at time t. The derivative of the velocity, $\frac{d}{dt}(\frac{dx}{dt})(t)$, is the *acceleration* of the object at time t.

The new function obtained by differentiating the derivative is called the *second derivative*. We can continue to take derivatives to obtain the third derivative, fourth derivative, and so on. Collectively, these are referred to as *higher-order derivatives*.

The notation for higher-order derivatives of $y = f(x)$ can be expressed in any of the following forms:

$$f''(x), f'''(x), f^{(4)}(x), \ldots, f^{(n)}(x)$$
$$y''(x), y'''(x), y^{(4)}(x), \ldots, y^{(n)}(x)$$
$$\frac{d^2y}{dx^2}(x), \frac{d^3y}{dx^3}(x), \frac{d^4y}{dx^4}(x), \ldots, \frac{d^ny}{dx^n}(x).$$

Thus, in the example above we may denote the acceleration by $x''(t)$ or $\frac{d^2x}{dt^2}(t)$.

Example. If $f(x) = x^2$, then $f'(x) = 2x$, and $f''(x) = (2x)' = 2$.

6. If $f(x) = 2x^2 - 3x + 1$, find $f''(x)$.